

AI Product Manager Copilot

Database Structure, Schemas, Use Case & Design Rationale

Dataset: 200 synthetic user feedback records | Workbook sheets: Users_Feedback and Theme_Summary

1. Database Use Case

The database is designed to provide structured customer-feedback data to the AI Product Manager Copilot. It enables the system to understand customer problems, identify common themes, analyze sentiment, estimate impact and reach, prioritize feature requests, generate PRDs, support roadmap planning, and produce product reports.

Core workflow: Customer Feedback → Ingestion → Sentiment Analysis → Theme Identification → Clustering → Impact/Reach Analysis → RICE Prioritization → Feature Prioritization → PRD Generation → Roadmap Planning → Reporting

2. Dataset Structure

Component	Purpose	Records / Fields
Users_Feedback	Main customer-feedback dataset	200 records × 22 columns
Theme_Summary	Aggregated analytical summary by theme and sentiment	10 themes × 4 columns

3. Users_Feedback Schema — 22 Columns

	Column	Purpose / Meaning
1	user_id	Unique identifier of the user.
2	feedback_id	Unique identifier of an individual feedback record.
3	user_name	Synthetic customer name used for demonstration.
4	feedback_timestamp	Date and time when feedback was submitted.
5	source	Origin of feedback, such as Email, Zendesk, or CRM.
6	plan	Customer subscription segment, such as Free, Starter, Pro, or Enterprise.
7	region	Geographical region associated with the feedback.
8	device	Platform/device context such as Web, Android, or iOS.
9	monthly_active_users	Activity measure used as product/customer context.
10	feedback_text	Original natural-language customer feedback.
11	sentiment	Detected sentiment: positive, neutral, or negative.
12	theme	Product problem area identified from feedback.
13	priority	Categorical product priority such as High, Medium, or Low.
14	urgency_score	Numerical measure of how urgently the problem needs attention.
15	impact_score	Estimated product/user impact of solving the problem.
16	confidence_score	Confidence in the assessment/classification.
17	estimated_effort	Estimated implementation effort.
18	affected_users	Estimated number of users affected by the issue.
19	reach_ratio	Relative reach/proportion of the affected user population.
20	rice_score	Calculated prioritization score based on RICE inputs.
21	feature_request	Product feature or improvement associated with the feedback.
22	status	Current product-development state, e.g. Planned, In Progress, or Completed.

4. Why Each Major Field Is Required

User and feedback identity: `user_id` identifies who gave the feedback, while `feedback_id` identifies which feedback record was given. This supports multiple feedback records per user and preserves traceability.

Time: `feedback_timestamp` enables trend analysis, recent-issue detection, and time-based reporting.

Source and customer context: `source`, `plan`, `region`, and `device` allow the PM to identify where problems originate and which customer segments or platforms are affected.

Activity context: `monthly_active_users` provides usage context that can help estimate the importance of a customer/product issue.

Raw feedback: `feedback_text` is the natural-language input analyzed by the AI agents. It is the basis for extracting sentiment, themes, and feature requests.

AI analysis: `sentiment` identifies positive/neutral/negative feedback, while `theme` converts unstructured comments into structured product problem areas.

Prioritization inputs: `priority`, `urgency_score`, `impact_score`, `confidence_score`, `estimated_effort`, `affected_users`, and `reach_ratio` provide the information needed for product prioritization.

RICE score: `rice_score` provides a quantitative basis for comparing initiatives rather than relying only on subjective judgment.

Product planning: `feature_request` identifies the product improvement, while `status` tracks its current development state.

5. RICE Prioritization

RICE is a prioritization framework used to compare product initiatives using four factors: Reach, Impact, Confidence, and Effort.

RICE = (Reach × Impact × Confidence) / Effort

In this dataset, the prioritization inputs are represented through fields such as `affected_users` / reach-related measures, `impact_score`, `confidence_score`, and `estimated_effort`. The resulting `rice_score` allows initiatives to be ranked quantitatively.

6. Theme_Summary Schema

Column	Meaning
<code>theme</code>	Product theme or problem area.
<code>negative</code>	Count of negative feedback records for the theme.
<code>neutral</code>	Count of neutral feedback records for the theme.
<code>positive</code>	Count of positive feedback records for the theme.

The `Theme_Summary` table is an analytical aggregation layer. It lets the Product Manager quickly see which product themes have the most negative, neutral, or positive feedback without reviewing every individual record.

7. Why Schemas Are Necessary

- **Consistency:** Every feedback record follows the same structure.
- **Validation:** The application can check whether expected fields and values are present.
- **Queryability:** Structured fields can be filtered and queried efficiently.
- **AI processing:** Agents know which fields contain raw feedback, sentiment, themes, and prioritization data.
- **Analytics:** Data can be grouped by theme, sentiment, plan, region, device, and other dimensions.
- **Scalability:** The same conceptual model can support hundreds, thousands, or millions of feedback records.
- **Traceability:** User and feedback identifiers preserve the relationship between individual feedback and analysis.

8. Excel Dataset vs Actual Application Database

The uploaded .xlsx file is technically a spreadsheet dataset, not a relational database. It is used as synthetic input data for development, testing, demonstration, and validation of the AI PM Copilot workflow.

The backend architecture uses SQLite for application persistence. The Excel dataset can be ingested into the application, while SQLite stores application-level information used by the running system.

SQLite table	Main fields	Purpose
feedback	id, filename, record count, created at	Tracks ingested feedback files and ingestion metadata.
features	id, name, rice score, status	Stores prioritized product features.
prds	id, title, content	Stores generated Product Requirement Documents.

9. Connection to the AI Agents

Agent	How the database supports it
Ingestion Agent	Reads the uploaded feedback dataset and extracts records, columns, and preview data.
Theme Agent	Analyzes feedback text to identify sentiment and product themes.
Clustering Agent	Groups related themes and feature requests into larger product initiatives.
Prioritization Agent	Uses reach, impact, confidence, and effort inputs to calculate/compare RICE scores.
PRD Agent	Uses prioritized feature information to generate a Product Requirements Document.
Roadmap Agent	Uses prioritized initiatives, dependencies, and status to support roadmap planning.
Reporting Agent	Aggregates feedback, themes, sentiment, and prioritization results into executive summaries and metrics.